

Estimating a Proportion with Starbursts

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

The Student's t distribution

If n is small (few observations) and σ is unknown, the standard normal distribution might not give a good approximation

If σ is known use standard normal:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

If σ is unknown in small samples use Student's t distribution

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n - 1).$$

Choosing the critical value

The critical value depends on two things:

- the confidence level
- the degrees of freedom, $n - 1$

For a 95% confidence interval, use $\alpha = 0.05$:

$$\text{critical value} = t_{0.025}(n - 1)$$

For a 90% confidence interval, use $\alpha = 0.10$:

$$\text{critical value} = t_{0.05}(n - 1)$$

Practice: t confidence interval

A sample of $n = 15$ observations from a normal population has

$$\bar{X} = 42.4, \quad s = 6.8.$$

Find a 95% confidence interval for μ

$$t_{0.025}(14) = 2.145$$

$$42.4 \pm 2.145 \left(\frac{6.8}{\sqrt{15}} \right) = 42.4 \pm 3.77$$

$$\boxed{(38.63, 46.17)}$$

Today's Experiment

Suppose a bag of Starbursts has four colors

We want to estimate:

p = the true proportion of red Starbursts

We can't count every candy produced by the factory, but we can sample

A bag should contain about 90 pink, red, orange, or yellow pieces

Sampling Procedure

Each group will take a random sample of Starbursts

For your sample, record:

- 1 Total number of candies sampled: n
- 2 Number of red candies: X
- 3 Sample proportion of red candies:

$$\hat{p} = \frac{X}{n}$$

Model the Distribution

Our random variable is:

X = number of red candies in a sample of size n

If each candy has probability p of being red, then:

$$X \sim \text{Binomial}(n, p)$$

Properties of X :

$$E[X] = np$$

$$\text{Var}(X) = np(1 - p)$$

Our Estimator and its Properties

Our “estimator” for p is the sample proportion:

$$\hat{p} = \frac{X}{n}$$

To use the CLT we need the mean and variance of $\hat{p}$

$$E[\hat{p}] = E\left[\frac{X}{n}\right] = \frac{1}{n}E[X] = \frac{np}{n} = p$$

$$Var(\hat{p}) = Var\left(\frac{X}{n}\right) = \frac{1}{n^2}Var(X) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$$

Standard Error

The standard deviation of the sampling distribution of $\hat{p}$ (standard error)

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

We do not know p , so we estimate it using $\hat{p}$:

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Using the CLT

For a large enough sample, the Central Limit Theorem implies:

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

Equivalently,

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \sim N(0, 1)$$

This lets us build a confidence interval for p

95% Confidence Interval

A 95% confidence interval for the true proportion of red candies is:

$$\hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Use $N(0, 1)$ here because σ^2 is known

estimate $\pm$ margin of error

where, margin of error = $1.96 \times \widehat{SE}(\hat{p})$

Data Table

Group	Sample size n	X (# red)	$\hat{p} = X/n$
1			
2			
3			
4			
5			

Each group gets a different estimate because of sampling variation

Combined Class Sample

After all groups report results, combine the data

X_{total} = total number of red candies

n_{total} = total number of candies sampled

The class estimate is:

$$\hat{p}_{\text{class}} = \frac{X_{\text{total}}}{n_{\text{total}}}$$

$$\hat{p}_{\text{class}} \pm 1.96 \sqrt{\frac{\hat{p}_{\text{class}}(1 - \hat{p}_{\text{class}})}{n_{\text{total}}}}$$

Example Calculation

Suppose the class samples $n = 80$ candies and finds $X = 28$ red candies.

$$\hat{p} = \frac{28}{80} = 0.35$$

The estimated standard error is:

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{0.35(1 - 0.35)}{80}} = 0.053$$

The 95% confidence interval is:

$$0.35 \pm 1.96(0.053) = (0.246, 0.454)$$

The confidence interval is:

$$(0.246, 0.454)$$

We are 95% confident that the true proportion of red Starbursts leaving the factory is between 24.6% and 45.4%.

What Happens if n Gets Larger?

Recall:

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

As n increases, the standard error decreases.

That means larger samples produce:

- 1 less sampling variation
- 2 narrower confidence intervals
- 3 more precise estimates